Lump Sum vs DCA Calculator

Personal Financial Scenario Report

Canada · CAD

fincalcsmart.com/lump-sum-vs-dca-calculator

EXECUTIVE SUMMARY

Lump Sum Leads

CA$193,484	CA$186,557	CA$6,927	62/100	Lump Sum Leads
Lump Sum FV	Monthly DCA FV	Lump Sum Advantage	Advantage Score Moderate	Comparison

FINAL VALUE COMPARISON (LUMP SUM VS DCA)

51% | 49%

Total projected: CA$380,041

Lump Sum FV: CA$193,484 (50.9%) Monthly DCA FV: CA$186,557 (49.1%)

AI-ASSISTED LUMP SUM VS DCA SUMMARY

Investing CA$50,000 at 7% (Monthly compounding) over 20 years: lump sum grows to CA$193,484 (gain: CA$143,484, +287%). Monthly strategy (CA$4,167/mo over 12 months) reaches CA$186,557 (gain: CA$136,557, +273%). Lump Sum Advantage Score: 62/100 (Moderate).

Lump sum advantage: CA$6,927 more at year 20 (significant difference). Milestone comparison -- at year 10: lump CA$98,358 vs DCA CA$94,836; at year 20: CA$193,484 vs CA$186,557; at year 30: CA$380,613 vs CA$366,986.

Lump sum investing tends to outperform gradual deployment in rising markets because the full principal compounds from day one. The monthly strategy (DCA) spreads market entry over 12 months, reducing timing risk but also reducing time-in-market for early tranches. Research consistently shows lump sum outperforms DCA approximately two-thirds of the time in equity markets. However, DCA is often the psychologically easier approach for risk-averse investors or when capital is received gradually (e.g., payroll). This model does not account for market volatility -- the advantage of DCA in volatile or declining markets can be larger than this comparison suggests.

INPUTS & ASSUMPTIONS

Total Investment	**CA$50,000**
Annual Return Rate	**7%**
Compounding Frequency	**Monthly**
Investment Horizon	**20 years**
DCA Spread Period	**12 months**
Monthly DCA Amount	**CA$4,166.67**
Region	**Canada**

DETAILED RESULTS

Lump Sum: Final Value	CA$193,484
Lump Sum: Gain	CA$143,484
DCA: Final Value	CA$186,557
DCA: Gain	CA$136,557
Lump Sum Advantage	CA$6,927
Difference Significant	Yes
Advantage Score	62/100 (Moderate)
At Year 10: Lump Sum	CA$98,358
At Year 10: DCA	CA$94,836
At Year 20: Lump Sum	CA$193,484
At Year 20: DCA	CA$186,557
At Year 30: Lump Sum	CA$380,613
At Year 30: DCA	CA$366,986

KEY DRIVERS / POSSIBLE ADJUSTMENTS

- The 7% return rate is the primary driver of the advantage gap. At higher return rates, lump sum benefits more from full compounding from day one -- the advantage widens with higher assumed returns.
- Spread period of 12 months directly determines the DCA disadvantage. A 3-month spread leaves most capital at risk for almost the full horizon; a 36-month spread more meaningfully reduces market-timing risk.
- Total investment horizon of 20 years amplifies any early-compounding advantage. At longer horizons, even a small initial advantage compounds into a larger absolute gap.

METHODOLOGY

What this calculator models:

- Lump sum: full principal P compounded for T months using the effective monthly rate derived from the selected compounding frequency: FV = P x (1 + r)^T.
- Monthly strategy (DCA): equal monthly instalments of totalAmount / spreadMonths, each compounded from its contribution month to end of horizon. FV = sum of monthly x (1+r)^(T-k) for k = 1 to spreadMonths.
- Advantage Score (0-100): derived from the percentage gain difference between strategies. Significant threshold: >5% absolute gain difference. ComparisonState "similar" when gainDelta is below the significance threshold.

What this calculator does not model:

- Market volatility -- lump sum outperforms in rising markets; DCA reduces loss in declining markets.
- Transaction costs, fund management fees (MER), or tax on capital gains.
- Dividend reinvestment or yield on fixed-income components.
- Tax-sheltered account rules (TFSA/RRSP in CA, 401k/IRA in US) that affect net return.
- Behavioural factors -- DCA often leads to better investor behaviour during market downturns.
- Sequencing: this model assumes the full lump sum is available immediately.

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